BUILDING A GOAL-DIRECTED AGENTIC SYSTEM USING AGNO

Smart Student Assistant Agent

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1. ABSTRACT

This project builds a goal-directed AI agent using the Agno framework. Unlike a normal Large Language Model (LLM) that only generates text from its training data, this agent can plan, use tools, observe results, and complete multi-step tasks. The agent was tested on the question: "What was Sudan's GDP in 2024 and multiply it by 2?" It successfully searched the web for fresh data, calculated the result using Python, and returned the correct answer.

2. INTRODUCTION AND PROBLEM STATEMENT

Large Language Models are powerful, but they have two main limitations:

1. They do not know fresh information after their training date.
2. They are not accurate in mathematical calculations.

The solution is an Agentic System: an LLM connected to tools, working in a loop: Think, Act, Observe, and Answer.

3. OBJECTIVES

1. Build an agent using Agno.
2. Connect the agent to two tools: Web Search and Python Calculation.
3. Test the agent on a multi-step task.
4. Compare Agno with LangGraph and CrewAI.

4. SYSTEM ARCHITECTURE AND TOOLS

Model: gemini-3.1-flash-lite via Google AI Studio

Tool 1: search_web - Searches the internet for fresh information. Uses ddgs library.

Tool 2: calculate - Calculates math expressions using Python.

Agent Code:

```
agent = Agent(
    model=OpenAIChat(id="gemini-3.1-flash-lite"),
    instructions="You are a Smart Student Assistant. When asked a question
that needs fresh data, first use search_web. When asked to calculate, use
calculate tool. Always explain your steps.",
    tools=[search_web, calculate],
    markdown=True
)
```

5. THE GOAL LOOP - HOW THE AGENT WORKS

1. **Think:** I need fresh data, I don't know GDP.
2. **Act:** Call search_web("Sudan GDP 2024")
3. **Observe:** Result: Sudan's GDP in 2024 was approximately 49.67 billion USD.
4. **Think:** Now I need to multiply 49.67 by 2.
5. **Act:** Call calculate("49.67 * 2")
6. **Observe:** Result: 99.34
7. **Answer:** Sudan's GDP in 2024 was 49.67 billion USD, multiplied by 2 = 99.34 billion USD.

6. RESULTS AND TESTING

Test 1 - Main Test:
Question: What was Sudan's GDP in 2024 and multiply it by 2?
Result: GDP = 49.67 billion USD, 49.67 * 2 = 99.34
Status: SUCCESS

Test 2 - Additional Test:
Question: What is 15 * 15? And what is 100 / 4?
Result: 15 * 15 = 225, 100 / 4 = 25
Status: SUCCESS

7. COMPARISON: AGNO VS LANGGRAPH VS CREWAI

Framework	Complexity / Learning Curve	Speed	Best Fit / Strengths
Agno	Very Easy / Low Learning Curve	Fast	Best for Single Agent, Easy Custom Tools
LangGraph	Hard / High Learning Curve	Medium	Best for Complex workflows with state control, Needs Graph design
CrewAI	Medium / Medium Learning Curve	Slow	Best for Team of agents, Needs Agents + Tasks + Crews

Why we chose Agno: For this project, we needed only one smart agent with two tools. Agno was the best choice because it is simple, fast, and easy to build custom tools.

8. CHALLENGES FACED

1. **Model Availability:** Google stopped old models like gemini-2.0-flash for new users. Solution: We used gemini-3.1-flash-lite.
2. **Agno Tool Error:** Built-in tools sent fields that Google API rejected. Solution: We built our own custom tools with @tool decorator.
3. **Search Failure:** Sometimes DuckDuckGo returns no results. Solution: We added try/except in search_web to handle errors.

9. CONCLUSION

This project proves that Agno can build a working goal-directed agent. The Smart Student Assistant successfully searched the web, calculated accurately, and explained its steps.

10. DELIVERABLES

- Working Jupyter Notebook (Google Colab) - Smart_Student_Assistant.ipynb
- This Report
- Presentation Slides